

Code Summary

I implemented the following steps in order to create a DistilBERT-based deep learning detector:

1. Step 1

First of all, I installed the required libraries necessary to run the transformer model.

The command is:

```
pip install transformers datasets torch scikit-learn tqdm
```

2. Step 2

Now, the model was trained on a single domain.

The command is:

```
python deep_learning_detector.py --mode single --domain phishing
```

3. Step 3

The model was then trained on all domains and was evaluated per-domain.

The command is:

```
python deep_learning_detector.py --mode all
```

4. Step 4

Finally, the model was evaluated in Zero-shot cross-domain. The model is trained on 6 domains and then it is tested on the 7th unseen domain i.e. Twitter Rumours.

The command is:

```
python deep_learning_detector.py --mode cross --held_out twitter_rumours
```

The entire code of deep learning detector model is present in the file “deep_learning_detector.py”.